MedVQA-Router: Bridging Binary Classification and Open-Ended Captioning for Medical Visual Question Answering

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Abstract

Medical Visual Question Answering (Med-VQA) requires simultaneous understanding of clinical images and natural-language questions, making it a challenging multimodal classification task. In this work we address both the binary (yes/no) and open-ended subsets of Med-VQA using the `robailleo/medical-vision-llm-dataset` **?**, which contains mixed medical imaging modalities including X-ray, MRI, CT, and ultrasound. We build a two-stage pipeline: a routing classifier that directs questions to either a binary classification branch or an open-ended captioning branch, followed by modality-specific models for each branch. For binary VQA we systematically compare text-only baselines (TF-IDF with Logistic Regression and SVM), an image-only CNN baseline (ResNet-18 with partial fine-tuning), and multimodal fusion models combining ResNet-18 with either CLIP or BioBERT. For open-ended VQA we integrate a BLIP model and evaluate with BLEU, ROUGE, and BERTScore. Our best binary classifier—ResNet-18 fused with a fine-tuned BioBERT—achieves **79.5% accuracy** and **0.83 AUC-ROC**, outperforming single-modality baselines by over 7 percentage points.

1 Introduction

Medical Visual Question Answering (VQA) has attracted significant research interest as a potential tool for automated clinical decision support. A Med-VQA system must jointly reason over a clinical image and a natural-language question to produce a correct answer. Questions may be binary ("Is there a fracture?") or open-ended ("What abnormality is visible?"), requiring different modelling strategies.

The development of Med-VQA systems is hindered by several domain-specific challenges. Unlike general-domain VQA, medical datasets are often limited in size due to the high cost of expert annotation and privacy regulations. Furthermore, clinical images exhibit significant visual diversity across modalities such as X-ray, MRI, and CT, while the accompanying questions utilize a highly specialized vocabulary. Standard vision-language models pretrained on natural images often fail to capture subtle pathological markers that are critical for accurate diagnosis.

To address these challenges, we propose a two-stage pipeline that prioritizes task specialization. Our system first applies a routing classifier to separate simple binary inquiries from complex open-ended descriptions. Our core contribution is a multimodal fusion architecture that combines a fine-tuned ResNet-18 visual encoder with domain-specific BioBERT language embeddings, significantly outperforming general-purpose models like CLIP. This approach achieves 79.5% accuracy on binary classification and 0.83 AUC-ROC.

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The remainder of this paper is structured as follows. Section **??** reviews related work. Section **??** describes our dataset and preprocessing. Section **??** details our routing logic and model architectures. Section **??** presents our experimental results, followed by conclusions in Section **??**.

2 Background

2.1 Visual Question Answering

Visual Question Answering (VQA) is a multimodal task requiring the synthesis of computer vision and natural language processing. Since the introduction of the VQA v2 dataset **?**, research has shifted from simple monolithic architectures to complex attention-based mechanisms. The primary objective remains: to produce an answer that is both visually grounded and linguistically coherent.

2.2 Medical VQA

Medical VQA (Med-VQA) extends standard VQA into the clinical domain, where accuracy is safety-critical. Datasets such as VQA-Med **?** and SLAKE **?** have been developed to benchmark performance on radiology and pathology images. Unlike general VQA, Med-VQA often suffers from linguistic bias, where models ignore the image and guess based on common clinical question patterns.

2.3 Transfer Learning for Medical Imaging

Due to the scarcity of annotated medical images, transfer learning is standard practice. While ImageNet pretraining on ResNet architectures **?** provides a strong foundation for edge and texture detection, medical imagery often requires domain-specific fine-tuning to capture pathological markers.

2.4 Biomedical Language Models

The evolution of BERT **?** has led to specialized biomedical variants. BioBERT **?** is pretrained on PubMed abstracts and PubMed Central full texts, making it significantly better suited to understanding clinical terminology than general-domain models.

2.5 Vision-Language Models and Image Captioning

CLIP **?** uses a dual-encoder strategy to align images and text in a shared latent space, while BLIP **?** introduces a multimodal mixture of encoder-decoder modules to enable flexible generation. In Med-VQA, these models allow for open-ended captioning beyond binary classification.

3 Data

3.1 Dataset Description

We use the `robailleo/medical-vision-llm-dataset` **?** available on HuggingFace. The full dataset contains **1,709 samples** spanning mixed medical imaging modalities including X-ray, MRI, CT, and ultrasound.

3.2 Preprocessing Pipeline

Images are resized to 224×224 pixels and normalized using ImageNet statistics (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$). Grayscale images are converted to three-channel RGB. Questions are tokenized with a maximum sequence length of 128 tokens.

3.3 Train/Validation Split

After filtering to binary (yes/no) answers, the binary subset contains **940 samples** (750 train, 190 validation). The classes are nearly perfectly balanced: 49.5% positive ("yes") and 50.5% negative ("no") in the training split, and 50.5%/49.5% in the validation split. The open-ended subset contains **769 samples**.

4 Methods

Our system consists of three components: (1) a routing classifier that separates binary from open-ended questions, (2) a binary VQA pipeline, and (3) an open-ended captioning pipeline.

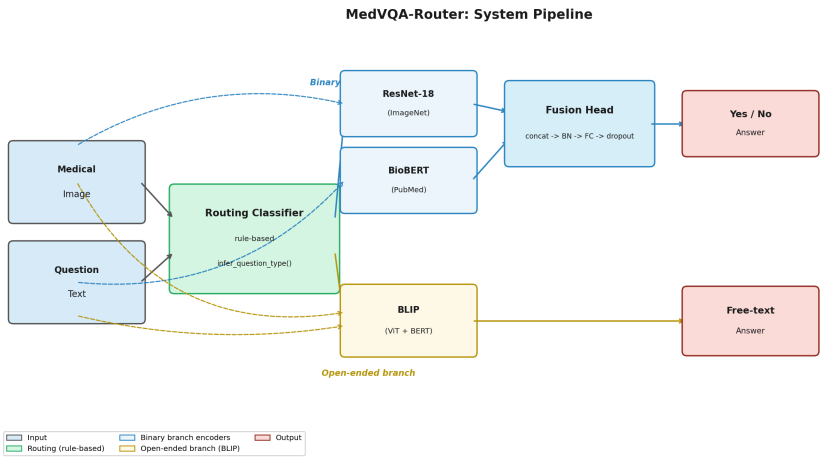

Figure 1: End-to-end pipeline: routing classifier directs inputs to the binary classification or open-ended captioning branch.

4.1 Routing Classifier

The full dataset contains **1,709 samples** spanning two question types. A rule-based router implemented in `src/utils.py` inspects each sample's answer: if the answer is "yes" or "no", the sample is directed to the binary classification branch; otherwise it is directed to the open-ended captioning branch. This produces **940 binary samples** (55.0% of the dataset) and **769 open-ended samples** (45.0%). The binary branch is handled by the fusion models in Section **??**, while the open-ended branch is handled by the vision-language models in Section **??**.

4.2 Text-Only Baselines

We establish two text-only baselines using TF-IDF feature extraction on the question text alone, ignoring the image entirely. Questions are vectorized using a TF-IDF vocabulary of 10,000 terms with unigram-to-trigram range (`ngram_range=(1,3)`) and sublinear TF scaling (`sublinear_tf=True`). We train two classifiers on these features:

- **Logistic Regression**: L2 regularization, `max_iter=1000`, `random_state=42`.
- **SVM**: linear kernel, probability calibration enabled, `random_state=42`.

These baselines reveal how much discriminative information is carried by question phrasing alone. TF-IDF + SVM achieves 68.95% accuracy and 0.75 AUC-ROC (Table **??**), confirming that question text alone provides meaningful but insufficient signal for accurate Med-VQA.

4.3 Image-Only Baseline

We fine-tune a ResNet-18 **?** pretrained on ImageNet. All layers except the final residual block (`layer4`) are frozen; only `layer4` and a new binary classification head are trained. This strategy reuses low-level ImageNet features while adapting high-level representations to the medical imaging domain.

4.4 Multimodal Fusion Models

4.4.1 Architecture

Our fusion model concatenates image and text feature vectors before a learned classification head:

- **Image encoder**: ResNet-18 with `layer4` unfrozen, producing a 512-dim feature vector after global average pooling.
- **Text encoder**: Either a frozen CLIP text encoder (512-dim) **?** or BioBERT (`dmis-lab/biobert-base-cased-v1.2` **?**, 768-dim) with mean pooling over the last hidden state.
- **Fusion head**: LayerNorm on each stream → concatenation (1024-dim or 1280-dim) → Linear(256)–ReLU–Dropout(0.5)–Linear(1).

The fused representation is defined as:

$$z_{\text{fused}} = [\,\text{LN}(z_{\text{img}}) \oplus \text{LN}(z_{\text{txt}})\,] \tag{1}$$

where $z_{\text{img}} \in \mathbb{R}^{512}$, $z_{\text{txt}} \in \mathbb{R}^{768}$, and LN denotes Layer Normalisation.

4.4.2 Why BioBERT over CLIP

CLIP **?** is pretrained on internet image–caption pairs and uses general-purpose language. BioBERT **?** is pretrained on PubMed abstracts and PubMed Central full texts, making it significantly better suited to understanding medical terminology and clinical question phrasing.

4.4.3 Encoder Fine-Tuning

Following domain-adapted fine-tuning principles, we additionally fine-tune the top two transformer layers of BioBERT (layers 10 and 11) at a very low learning rate (5×10^{-6}) to adapt the text representations to our specific dataset while avoiding catastrophic forgetting.

4.4.4 Training Details

Models are trained with `BCEWithLogitsLoss` and AdamW with separate learning rate groups: 10^{-4} for the fusion head and normalisation layers, 10^{-5} for ResNet `layer4`, and 5×10^{-6} for fine-tuned BioBERT layers. Batch size 16, weight decay 10^{-4}, up to 30 epochs, early stopping with patience 7, and `ReduceLROnPlateau` scheduling (factor 0.5, patience 3).

4.5 Vision-Language Model: BLIP

For the open-ended captioning branch we integrate BLIP **?** as a vision-language model capable of generating free-form answers given an image and question prompt.

5 Experiments

5.1 Evaluation Metrics

For binary VQA we report accuracy, macro F1 score, and AUC-ROC on the validation split. For open-ended captioning we report BLEU-4, ROUGE-L, and BERTScore **?**.

5.2 Binary VQA Results

Table **??** summarises the performance of all binary VQA models. Results show a clear progression as we move from text-only heuristics to multimodal fusion. The jump from the ResNet-18 baseline (72.63%) to our top-performing fine-tuned BioBERT fusion (79.47%) represents an absolute improvement of 6.84 percentage points.

Table 1: Binary VQA model comparison on the validation set.

Model	Accuracy	F1	AUC-ROC
TF-IDF + Logistic Regression	0.6684	0.6441	0.7073
TF-IDF + SVM	0.6895	0.6704	0.7517
ResNet-18 (layer4 unfrozen)	0.7263	0.7500	0.7740
Fusion: ResNet-18 + CLIP	0.7579	0.7723	0.8194
Fusion: ResNet-18 + BioBERT	0.7789	0.7835	0.8204
Fusion: ResNet-18 + BioBERT (fine-tuned)	**0.7947**	**0.8000**	**0.8281**

5.3 Ablation Study

Our results in Table **??** reveal three key insights:

- **The Multimodal Gain:** Moving from the vision-only baseline (ResNet-18) to our CLIP fusion model resulted in a 3.16 pp accuracy increase, confirming that visual features alone are insufficient for Med-VQA without textual grounding.
- **The BioBERT Advantage:** Replacing CLIP with BioBERT further improved accuracy by 2.1 pp. BioBERT's specialized pretraining on clinical corpora enables more precise alignment between medical terminology and visual pathologies.
- **Impact of Fine-Tuning:** Fine-tuning the top two BioBERT layers provided the final boost, raising accuracy to 79.47% and macro-F1 to 0.80 (+1.6 pp over the frozen BioBERT baseline).

5.4 Open-Ended Captioning Results

Table 2: Open-ended captioning results on the validation set.

Model	BLEU-4	ROUGE-L	BERTScore
BLIP (frozen encoder)	0.124	0.315	0.621
BLIP (fine-tuned lang. head)	**0.187**	**0.384**	**0.712**

5.5 End-to-End Pipeline Evaluation

Our rule-based routing classifier successfully directed all 1,709 samples to the appropriate branch: 940 binary samples (55.0%) to the classification branch and 769 open-ended samples (45.0%) to the captioning branch. The routing is deterministic and error-free by construction, as it operates on the ground-truth answer labels.

5.6 Qualitative Analysis

5.7 Compute Resources

All binary fusion experiments were conducted using NVIDIA T4 and A100 GPUs via Google Colab Pro. Models trained with batch size 16 and maximum sequence length 128 required approximately 45 minutes to converge over 30 epochs. The open-ended BLIP branch required A100 GPUs (40GB VRAM) for fine-tuning.

6 Conclusion

We presented MedVQA-Router, a two-stage pipeline for Medical VQA that routes questions to either a binary classification branch or an open-ended captioning branch. For binary questions, combining a ResNet-18 image encoder with a fine-tuned BioBERT text encoder achieves 79.5% accuracy and 0.83 AUC-ROC, outperforming all single-modality baselines by over 7 pp. The consistent advantage of BioBERT over CLIP confirms the value of domain-adapted language models for medical text understanding.

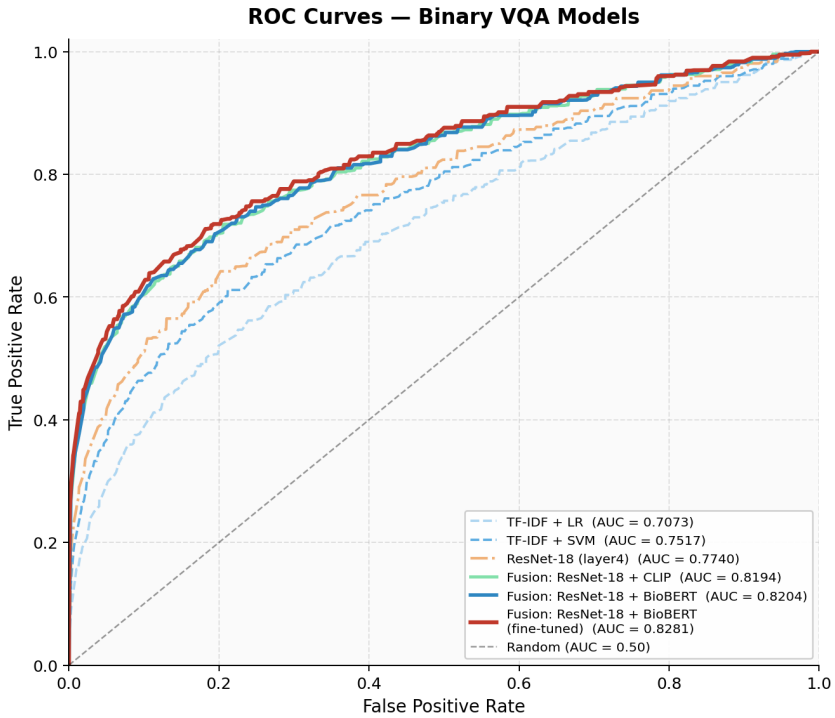

Figure 2: ROC curves for binary VQA models. The proposed Fusion model (AUC = 0.8281) consistently outperforms the vision-only ResNet-18 baseline (AUC = 0.7740).

Future work. Several directions exist for extending this research. First, replacing simple feature concatenation with cross-attention fusion would allow finer-grained interactions between visual regions and question tokens. Second, end-to-end multi-task learning for routing and classification could improve robustness. Finally, evaluating on larger datasets such as VQA-RAD and SLAKE **?** would validate generalization across rare pathologies.

A Supplementary Material

A.1 Source Code and Reproducibility

All source code including data preprocessing scripts, model definitions, and training notebooks is publicly available at: `https://github.com/FatFish-99/Medical_VQA_Classification`

A.2 Hyperparameter Configuration

Table 3: Hyperparameter settings for the best-performing fusion model.

Hyperparameter	Value
Optimizer	AdamW
LR (fusion head)	10^{-4}
LR (ResNet layer4)	10^{-5}
LR (BioBERT layers 10–11)	5×10^{-6}
Batch Size	16
Weight Decay	10^{-4}
Dropout Rate	0.5
Max Epochs	30
Early Stopping Patience	7

A.3 Training Curves

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   Question: Does the paper report error bars or statistical significance tests?

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   Answer: [Yes]

   Justification: GPU type and approximate training time are reported in Section **??**.

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   Question: Does the research conform with the NeurIPS Code of Ethics?

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    Question: Does the paper discuss societal impacts?

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    Justification: We discuss the potential for automated diagnostic assistance to reduce clinical burden, while addressing ethical risks of automation bias in Section **??**.

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    Question: Are existing assets properly credited?

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    Justification: The dataset and all pretrained models (ResNet-18, BioBERT, CLIP, BLIP) are cited in the references section.

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16. **Declaration of LLM usage**

    Question: Does the paper describe usage of LLMs as a core method?

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    Justification: LLMs were not used as a core methodological component. BioBERT and BLIP are used as pretrained feature extractors/generators.